

Remote sensing compatible snow loss modelling for PV power simulations

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ABSTRACT

In cold-climate regions, snow can reduce photovoltaic (PV) system power output by up to 100% during individual months and as much as 34 % annually, making accurate snow loss modeling essential for reliable PV power generation simulations. To address this, we present a generalizable snow loss model, inspired by the Marion model and designed to estimate hourly snow-induced PV power losses using remote-sensing derived PV system data, including aerial imagery, Light Detection and Ranging (LiDAR), and satellite-derived irradiance and weather data. The model is integrated into a simplified gridded PV simulation framework that reduces the computational time by 99% while matching the accuracy of the standard IV-model. The snow loss model shows consistent improvements in coefficient of determination, mean absolute error and root mean squared error, and results on par with previous snow loss studies. When incorporated into a full remote sensing-based PV power simulation pipeline, the resulting average percentage root mean square error was 5.7 % when simulating the hourly power output across 40 systems. This demonstrates that individual PV power generation from multiple distributed PV systems can be assessed at scale in cold climates, even without access to system specific technical data.

1. Introduction

By the end of 2024, global cumulative PV capacity had surpassed 2.2 TW, up from 1.6 TW in 2023, resulting from the commissioning of over 600 GW of new PV systems [1]. The technical simplicity, scalability, and modularity of PV technology have led to low unit costs [2] and investment risks [3,4], enabling a large share of small-scale distributed systems, which currently account for approximately 45 % of the global market [1]. Consequently, a significant portion of PV power is consumed on-site, “behind the meter” [5], and cannot be measured by system operators such as Transmission System Operators (TSO) or local Distribution System Operators (DSO) [6]. From the electricity system perspective, this self-consumed power is perceived as a reduction in power demand and is generally not included in local or national formal

statistics.

The variability and uncertainty of weather conditions affecting PV power output is a major challenge for integrating high shares of PV [7] and must be addressed to maintain grid stability [8,9]. Accurate forecasts of both generated and self-consumed distributed PV power at local [10,11] and regional [12–15] scales are crucial to maintain grid stability, reduce system costs [6], and minimize curtailment [16].

Simulating power generation from distributed PV within different time horizons — historical simulations, real-time and long-term forecasting — is an active research topic [6–18]. PV power simulating/forecasting approaches are typically categorized into four major groups [6–18] (1) persistence models, (2) physical/parametric methods, (3) statistical/data-driven methods, or (4) hybrid models. Persistence techniques are generally limited to very short-term predictions [17–19],

Abbreviations: AC, Alternating Current; CNN, Convolutional Neural Network; Cfb, temperate oceanic climate Köppen-Geiger climate zone; DC, Direct Current; Dfb, warm-summer humid continental Köppen-Geiger climate zone; Dfc, subarctic Köppen-Geiger climate zone; DSO, Distribution System Operator; EPI, Energy Performance Index; ERA5, fifth generation ECMWF atmospheric reanalysis of the global climate; IV, current-voltage; LiDAR, Light Detection and Ranging; MAE, Mean Absolute Error; NOCT, Nominal Operating Cell Temperature; NWP, Numerical Weather Predictions; POA, Plane Of Array irradiance; pRMSE, percentage Root Mean Square Error; PV, Photovoltaics; QCPV, Quality Control for PV reference systems; R^2 , Coefficient of determination; RMSE, Root Mean Square Error; SARAH-3, Surface Radiation Data Set – Heliosat – Edition 3; SES, Solar Energy Systems; SMHI, Swedish Meteorological and Hydrological Institute; STC, Standard Test Condition; TSO, Transmission System Operator.

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while physical models often struggle to simulate aggregated distributed PV due to incomplete or inaccessible technical data about the PV systems — usually unknown orientation (tilt and azimuth) [20] — or lack of sufficient spatial and/or temporal resolution Numerical Weather Predictions (NWP) [6]. In contrast, statistical models can leverage historical patterns without system-specific information, though they require large historical datasets for training and evaluation [6].

Various strategies have been proposed to estimate or account for the missing orientation of distributed PV systems in large-scale simulations. Lately, bottom-up physical models for wide-area PV power generation have been proposed, leveraging remote sensing techniques to collect data on individual distributed PV systems [21–23]. This study builds upon that approach by using the *Alfröduell* remote sensing pipeline, which incorporates several different computational algorithms or calculation models developed in different studies [21,24–27]. Fig. 1 schematically illustrates the different model steps of *Alfröduell* and lists their origin.

The *Alfröduell* pipeline enables simulation of individual PV systems across a geographical area without manually collecting system-specific data. Unlike statistical upscaling methods, *Alfröduell* produces physically based PV simulations from actual distributed systems that can be aggregated at scales ranging from small clusters to larger regions, such as municipalities, depending on the analysis needs.

Disregarding solar irradiance fluctuation attributed to the earth's orbit around the sun, which can be accurately described by physical equations, solar irradiance is mainly influenced by the presence of clouds, whose stochastic presence blocks the rays of the sun. Furthermore, many distributed PV installations are affected by irradiance losses due to shadows from nearby obstacles, such as trees or buildings [28,29], which lead to PV power simulation errors [21]. Special cases of shadowing are the accumulation of dirt or snow on the PV modules. The latter — usually referred to as snow losses — can in some locations lead

to a complete loss of PV power generation during certain months [30–34] and up to 34 % losses on annual basis for obstructed modules [35]. Importantly, snow shading losses on system level are also present in both the warmer and colder parts of the continental climate zone, like southern Canada (Dfb) [36], northern United States (Dfb) [37], the Rocky Mountains in the United States (Dsa) [31,32], central Norway (Dfb & Dfc) [34], northern and central Sweden (Dfc & Dfb) [38] and northern Japan (Dfb) [39], where annual losses between 3–20 % have been reported (letters within the parentheses refer to the detailed Köppen–Geiger climate classification [40]). For the temperate climate zone like western Norway (Cfb) [34] and southern Sweden (Cfb), snow losses typically range 2–6 %. As PV penetration in high-latitude regions increases, so does the need for accurate physical PV power simulations and forecasts that include snow loss models.

Snow cover affects PV modules similarly to complete or partial shading from surrounding objects, and the shading response of a PV system partly covered by snow depends on various factors, including module orientation [37], module type (crystalline silicon or thin film), type of snow [35,41,42], snow shading shape, and technical aspects of the system, such as the substring configuration, type of inverter and the use of optimizers. Estimating the snow cover for PV systems by deriving physical and empirical models is complex also because of various factors that affect the snow accumulation, including weather conditions — like irradiation [43], temperature and wind speed [44,45] — and technical aspects of the PV system — such as module mounting [37], roof or ground configuration [31,37], tilt [31,36,37,41], tracking [32] and module technology [44,46]. The above-mentioned factors also have different effects on the natural snow-clearing processes, which primarily are melting and sliding [30–34]. These processes typically occur at temperatures above 0 °C [35,46] although both sliding [30,42] and melting [43] can happen at ambient temperatures below freezing.

For snow clearance through sliding, several PV system design

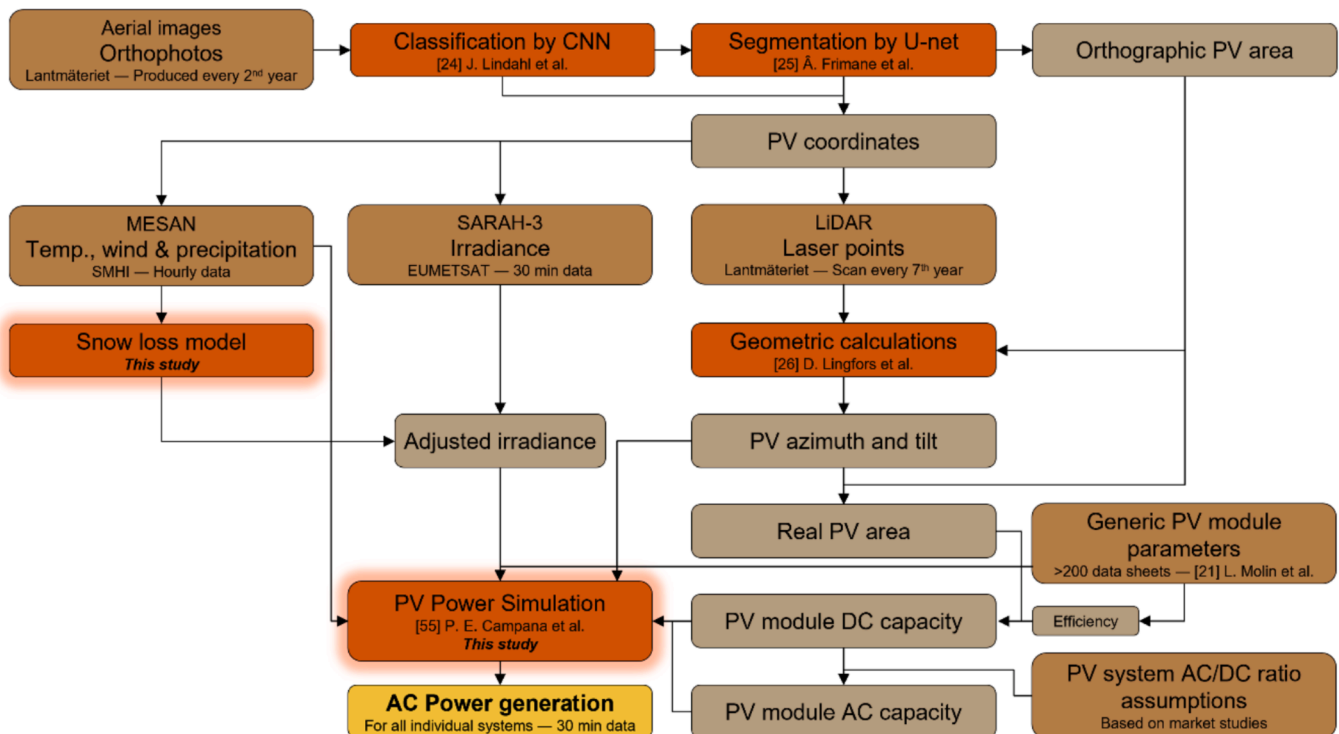


Fig. 1. Schematic flowchart of *Alfröduell*'s model steps for generating input data for a physical PV power simulation, which estimates 30 min PV power generation from all PV systems across a geographical area. Brown boxes indicate input data, its source, and time resolution. Orange boxes represent computational models or calculations, with references to relevant studies describing them. Arrows illustrate data flow between computational steps, with the resulting intermediate outputs shown in light brown boxes and the final simulated PV power generation data in the yellow box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

elements, such as tilt angle [30,41] and obstacles like gaps between modules and snow fall protection measures [44], can impact snow sliding and contribute to ground or roof interference [31,37]. Furthermore, friction and adhesion between the PV modules and the snow are critical, which depends on the type of PV module — including the type of glass and the presence of a frame [46] — as well as the composition of the snow [35,43–45].

Numerous attempts have been made to model snow losses, and based on prediction resolution and methodology, these models can be divided into two categories: (1) direct electricity reduction models, known as *power models*, and (2) *snow cover models* [35]. Power models estimate electricity losses by using weather data and system characteristics, with predictions ranging from daily to annual time steps [35]. In contrast, snow cover models focus on simulating the snow clearing processes, allowing for the prediction of snow cover on PV modules at a (sub) hourly resolution. Table 3 summarizes existing snow loss models.

However, most existing snow loss models require detailed system-specific input data, which limits their applicability in large-scale simulations of distributed PV. Therefore, this study develops a snow loss model specifically designed for integration with physically driven PV power simulations at system level and hourly resolution, utilizing only remote sensing data across a large number of PV systems. In this study, the snow loss model is integrated into the physical PV power simulation framework of the *Alfröddull* remote sensing pipeline but can also be used in other similar applications.

2. Data and methodology

2.1. Weather, climate and irradiance data

The data used in this study includes weather data from the MESAN model [47], developed by the Swedish Meteorological and Hydrological Institute (SMHI), and irradiance data from CM SAF SARA-3 [48] (©2024 EUMETSAT). SARA-3 provides instantaneous irradiance (W/m^2) at 30-minute intervals. To align with the hourly MESAN weather data and measured PV electricity generation of the reference systems described in Section 2.2, the irradiance data were converted to hourly irradiation values (Wh/m^2). Because the irradiance values are instantaneous, a simple hourly average would not match the hourly average

PV power generation data. Instead, a weighted average was applied to better align the datasets. For example, irradiation from 8:00 to 9:00 is computed as: $8:00 \times \frac{1}{4} + 8:30 \times \frac{1}{2} + 9:00 \times \frac{1}{4}$, matching the PV power generation average for the 8:00–9:00 interval.

The MESAN dataset [47] provides hourly data on ambient temperature ($^{\circ}\text{C}$), wind speed (m/s), and snowfall rate (cm/h). Snow depth data are obtained from ERA5 [49]. Fig. 2 presents the monthly average snow depth over seven winters for systems in the validation set, clearly indicating that snow-related losses are relevant in Sweden.

2.2. The reference PV systems

To evaluate the PV power simulation model (further described in Section 2.4) including a snow loss model (Section 2.6), PV power generation data from multiple PV systems with known characteristics were used. The original set of 40 reference systems came from two sources: seven systems from the Swedish aggregator CheckWatt — provided to a previous study [21] with 1-minute time resolution, and 33 systems from SolarEdge's public open monitoring platform [50] with 15-minute resolution. These datasets were aggregated to hourly resolution to enable evaluation of all reference systems at a common temporal resolution, consistent with the weather data resolution. For these reference systems, input data were collected to support the PV power simulations, including DC and AC capacities, geographic coordinates (WGS84), orientation (tilt and azimuth), and PV module characteristics.

Accurate orientation — both azimuth and tilt — is critical for estimating plane-of-array irradiance and for modeling snow shedding behavior, as the clearing rate is influenced by the sine of the tilt angle. For example, Van Noord et al. [38] demonstrated that a 10° tilt error (in a system with a true tilt of 25°) can result in a 25 % difference in the time required for snow to clear from a PV module.

Light Detection and Ranging (LiDAR)-derived system orientations have demonstrated higher accuracy than those reported by system owners, which often include uncertainties [21,29]. To validate the accuracy of the orientation data provided by the system owners in this study, both azimuth and tilt were assessed using the LiDAR-based method described in [26], which has demonstrated a 95 % success rate for azimuth estimation.

To further ensure the reliability of the reference system orientations,

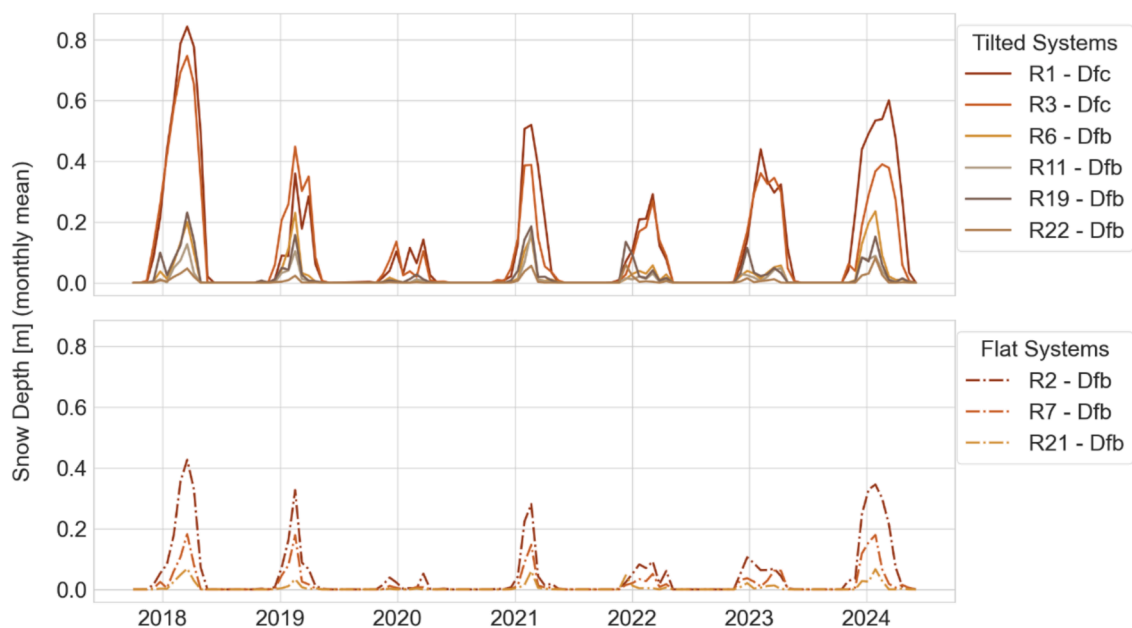


Fig. 2. Monthly mean snow depth across seven winters of data from ERA5 [49] for the validation set of reference systems. The top graph shows snow depth at the locations of the tilted systems, and the bottom at those of the flat systems.

additional verification steps were conducted. Azimuth values were cross-validated using the Azimuth Measurement plugin in QGIS, confirming agreement within $\pm 3^\circ$ for all systems with adequate LiDAR coverage. Tilt angles were verified through a combination of Google Street View inspection and trigonometric calculations, with LiDAR-derived tilt estimates accurate within $\pm 5^\circ$ for systems supported by high-quality imagery.

Quality control of the generation data was performed using the QCPV (Quality Control for PV reference systems) method described in [51], with adaptations from [52]. Although QCPV includes a system orientation parametrization step, it was omitted here due to the use of validated LiDAR-derived orientation data of the reference systems. Tests 1 through 7 from the QCPV protocol were applied, but Test 4 — daily energy ratio test — was modified to avoid flagging days with potential snow losses. Snow presence was identified using snow depth data from the ERA5-Land reanalysis dataset [49], obtained via the Copernicus Climate Change Service [53]. This ensured that low generation caused by snow cover was not mistaken for system faults. Nonetheless, manual inspection was required to identify and exclude periods of obvious system malfunction.

Shading was identified by comparing the hourly profile (hours 1–24) of measured and simulated PV power generation. Heavily shaded reference systems were excluded from the snow loss model development. Large, regular deviations under clear-sky conditions — determined using the CAL (cloud albedo) variable from SARAH-3 irradiance data — were attributed to shading. Suspected cases were verified through Google Maps and Street View to confirm the presence of obstructions such as buildings or trees. After quality control and the exclusion of 15 shaded systems, five CheckWatt systems and 20 Solar-Edge systems remained for the snow model development and evaluation.

These remaining reference PV systems exhibited a wide range of tilts. For analytical purposes, they were divided into two groups: (1) systems with tilt angles $\leq 12^\circ$, typically found on flat roofs, and (2) systems with tilt angles $> 12^\circ$, generally associated with tilted roof buildings. In terms of geographical distribution, 21 of the systems were found to be located in the warm-summer humid continental zone (Dfb), while the remaining four are situated in the subarctic zone (Dfc), according to the Köppen–Geiger climate classification [40], as illustrated in Fig. 3.

To enable evaluation of model performance on an independent set of systems not used during development, the 25 reference PV systems were divided into two sets: one consisting of 16 systems for snow loss model development and one consisting of 9 systems for validation. To maintain representativeness, both sets included a balanced mix of flat and tilted roofs, broad geographical distribution, and consistent coverage of climate zones (Dfc and Dfb).

The reference systems used in the study with their system characteristics are listed in Table 1, and the geographical distribution is shown in Fig. 3.

The development set includes data from the winters of 2021/2022, 2022/2023, and 2023/2024. Although some systems have longer time series, the same three winters were used across all development systems to ensure consistency and equal weighting. For validation systems, all available winters were used to test the model on independent data.

2.3. The evaluation metrics

To evaluate the accuracy of the developed snow-loss model, a set of evaluation metrics is needed. The evaluation metrics, and their corresponding skill score functions, used in this study are detailed in Table 2.

In the evaluation matrix formulas, y represents simulated PV power generation, x denotes measured PV power generation and $\bar{x}$ is the overall mean of the measured PV power generation. In the skill score functions lower case *sm* refers to snow model simulations and *b* to baseline PV power simulations without a snow loss model.

The coefficient of determination (R^2) is used to evaluate how well the

model explains variance in observed values, indicating overall fit. Root Mean Squared Error (RMSE) is included for its sensitivity to large errors, helping to identify significant discrepancies in snow loss modeling. To assess overall error magnitude, Mean Absolute Error (MAE) is preferred due to its robustness and lower sensitivity to outliers compared to RMSE. Accordingly, MAE is used as the target metric for model parameter optimization.

2.4. The baseline PV power simulation model

Various physical models are available for simulating PV system power output, including the widely used method of modelling the PV current–voltage (IV) curve using the one-diode equation. As a simplified alternative to the detailed IV curve model, a gridded simulation model, originally stemming from the linear physical PV power generation model presented in [54], has been developed by Campana et al. [55] for solar PV applications in the Nordic countries. The Appendix compares the simplified gridded simulation model to the detailed IV-curve model. The output differences are minor, while the gridded model requires fewer input parameters and offers 99 % faster computation — both key advantages for remote sensing. It is therefore chosen as the foundation for the *Alfrödu* remote sensing pipeline and the snow loss model development. The Appendix also outlines improvements to previously published versions of the gridded model [21,55], resulting in an updated equation for simulating PV power generation per installed module capacity ($P_{DC/kW}$). This updated formulation is used throughout the study and referred to as the baseline PV power simulation model. The equation, same as Eq. (A.3), is as follows:

$$P_{DC/kW} = \frac{S \cdot \eta \cdot \left(1 + \frac{\mu_{Voc}}{V_{mp}} \left(T_a - T_{STC} + \frac{G_{g,t}}{800} (T_{NOCTa} - 20) \right) \left(1 - \frac{\eta}{\tau\alpha} \right) \frac{9.5}{5.7 + 3.8 WS_{PV}} \right)}{G_{STC} \cdot \eta}, \quad (1)$$

where S is the PV system effective absorbed global tilted irradiance (W/m^2), η is the efficiency of the PV module at standard test conditions (STC), μ_{Voc} is the open circuit voltage temperature coefficient ($V/^\circ C$), V_{mp} is the voltage at maximum power point (V), T_a is the ambient temperature ($^\circ C$), T_{STC} is the temperature during STC ($25^\circ C$), $G_{g,t}$ is the incoming global tilted irradiance (W/m^2), T_{NOCTa} is the NOCT temperature adjusted based on the system mounting, $\tau\alpha$ is the transmittance absorptance product, WS_{PV} is the wind speed (m/s) at the height of PV system, and G_{STC} is the global tilted irradiance at STC, which is equal to $1\,000\,W/m^2$.

Fig. 4 shows a scatterplot comparing measured and simulated PV electricity generation by the baseline PV power simulation model for two of the reference systems of this study.

The close alignment between the regression line and the perfect fit suggests a good overall match. However, a noticeable cluster of discrepancies occurs along the y-axis, where the model simulates generation despite measured values being near zero. These points predominantly fall between November and April, suggesting snow coverage as a likely cause. Including a snow loss model has hence the potential to improve the simulation accuracy in snowy climates by accounting for seasonal deviations and correcting spurious low-generation points.

2.5. PV snow loss modelling

To identify a suitable snow loss model a literature review was conducted, summarized in Table 3. The selection focused on models with hourly resolution and minimal data requirements, aligning with the goals of this study to incorporate a snow loss model into the *Alfrödu* PV power simulation tool.

Models without hourly resolution were excluded based on the intended scope and the first-principles models were dismissed due to

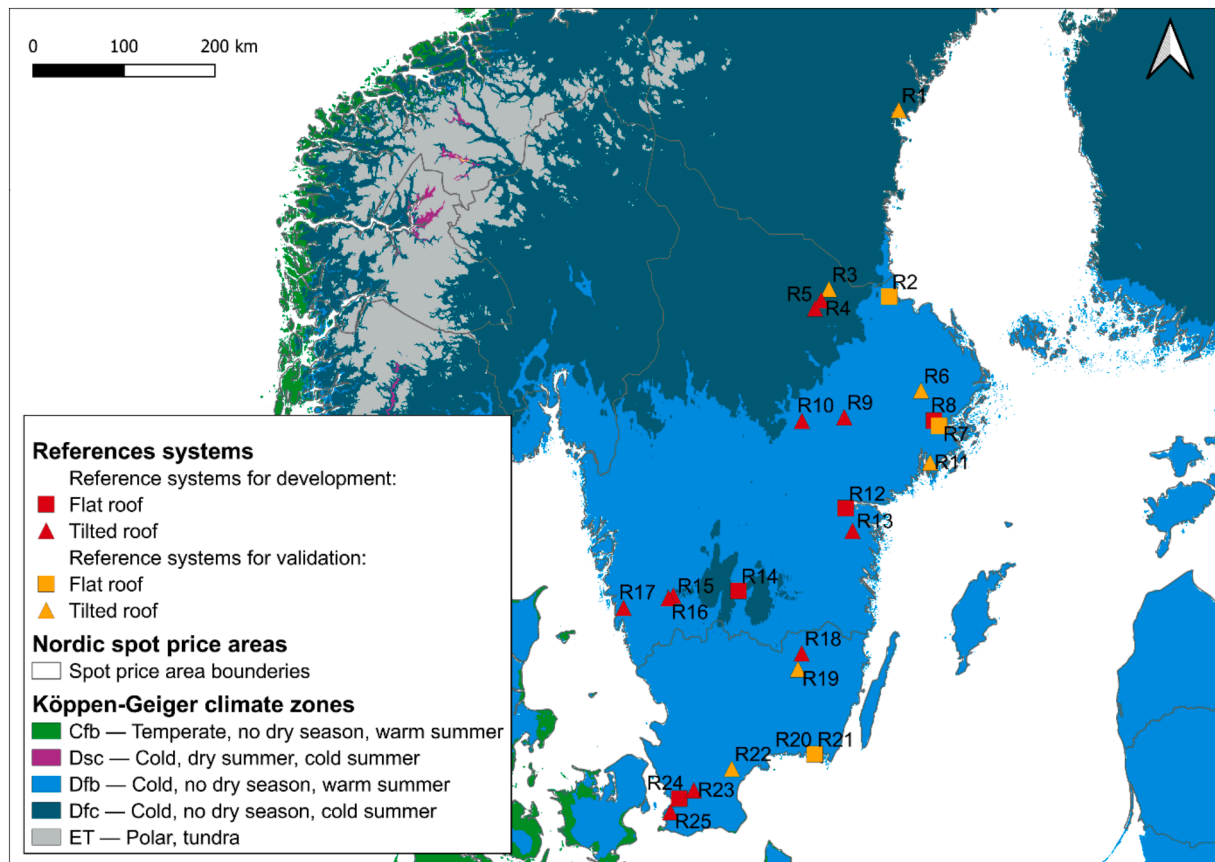


Fig. 3. The geographical location of the reference systems used in the study, where the color of the map shows the Köppen-Geiger climate zones [40]. Reference systems used for model development are shown in red and for evaluation in yellow. Flat roof systems are indicated by squares while tilted roofs are indicated by triangles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Summary of reference PV system data. A lowercase number indicates multiple of azimuths/tilts of that subsystem. A lowercase number denotes multiple clusters of modules with the same azimuths/tilts. For instance, 36₃, 39 in the tilt column indicate three clusters of modules with a tilt of 36° and one cluster with 39°. The “set” column refers to the application of the reference system, whether it is used for model development or model validation.

System	Location	Climate Zone	DC Capacity [kW]	AC Capacity [kW]	Tilt(s)	Azimuth(s)	Subsystems	Roof type	Set
R1	Timrå	Dfc	17.6	17.0	28, 23, 23	49, 139, -41	3	Tilted	Val
R2	Gävle	Dfb	16.7	17.0	28 ₂ , 7, 6	-28, 62 ₂ , -28	4	Flat	Val
R3	Falun	Dfc	19.5	16.4	36 ₃ , 39	39, -50, -50, 40	4	Tilted	Val
R4	Falun	Dfc	9.3	7.4	43, 20	-41, 49	2	Tilted	Dev
R5	Falun	Dfc	17.2	15.0	45	55	1	Tilted	Dev
R6	Knivsta	Dfb	8.0	16.4	21, 22	-12, 78	2	Tilted	Val
R7	Solna	Dfb	140.4	155.0	10 ₁₁	58 ₃ , 11 ₃ , 43 ₃ , -2 ₃	11	Flat	Val
R8	Sollentuna	Dfb	41.9	50.0	10 ₅	78 ₃ , 91, 1	5	Flat	Dev
R9	Köping	Dfb	13.5	14.0	15, 33	-20	2	Tilted	Dev
R10	Lindesberg	Dfb	44.0	50.0	15 ₂	10 ₂	2	Tilted	Dev
R11	Nynäshamn	Dfb	9.2	15.0	25	6	1	Tilted	Val
R12	Norrköping	Dfb	55.2	50.0	12 ₂	-14 ₂	2	Flat	Dev
R13	Valdemarsvik	Dfb	32.5	34.0	22, 21	-46	2	Tilted	Dev
R14	Jönköping	Dfb	29.1	25.0	63	4, -87, -42	3	Flat	Dev
R15	Borås	Dfb	11.9	12.5	14	-32	1	Tilted	Dev
R16	Borås	Dfb	13.6	17.0	27	-29	1	Tilted	Dev
R17	Möndal	Dfb	12.8	15.0	21, 21	3, -3	2	Tilted	Dev
R18	Uppvidinge	Dfb	10.1	9.2	27	6	1	Tilted	Dev
R19	Uppvidinge	Dfb	11.4	17.0	22	53	1	Tilted	Val
R20	Karlskrona	Dfb	33.0	30.0	0	-180	1	Flat	Dev
R21	Karlskrona	Dfb	33.0	30.0	0	-180	1	Flat	Val
R22	Kristianstad	Dfb	9.6	17.0	44, 8	-2	2	Tilted	Val
R23	Eslöv	Dfb	16.1	17.0	26, 10, 10	-8, -98, 82	3	Tilted	Dev
R24	Lund	Dfb	77.7	60.0	104	126, -54, 134, -46	4	Flat	Dev
R25	Malmö	Dfb	7.0	7.0	22	18	1	Tilted	Dev

Table 2

Definition of evaluation metrics and their corresponding skill score functions.

Evaluation metric	Definition	Skill score function
Coefficient of determination	$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - x_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$	$R_{skill\ score}^2 = \frac{R_{sm}^2}{R_b^2} - 1$
Root Mean Square Error	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2}$	$RMSE_{skill\ score} = 1 - \frac{RMSE_{sm}}{RMSE_b}$
Mean Absolute Error	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - x_i $	$MAE_{skill\ score} = 1 - \frac{MAE_{sm}}{MAE_b}$

their reliance on detailed, site-specific data which is often missing. The model by Gun et al. [64] (SunPower model) was also excluded due to its complexity and focus on ground-mounted systems. Based on these criteria, a modified version [34] of the original Marion model [61] was selected for implementation. Study [65] showed that this version yielded lower errors for both residential and commercial PV systems compared to the original implementation. Hereafter, this version is referred to as the *Modified Marion* model.

2.5.1. The Modified Marion snow loss model

The *Modified Marion* model applies a threshold-based method with hourly resolution. It assumes that PV modules are fully snow-covered after snowfall, triggered when hourly snowfall exceeds a threshold $TH_{snowfall}$. Snow accumulation is tracked using hourly snowfall data. During hours without new snowfall, snow is assumed to melt and slide off the modules, provided the sliding condition is met:

$$T_a > \frac{G_{g,t}}{m}, \quad (2)$$

where T_a is the ambient temperature and m represents an empirically determined factor valued at $-80 \text{ W/}^\circ\text{Cm}^2$ [51]. If the condition in Equation (2) is met during any given hour, snow will slide, given by the sliding fraction, D , of the total row height:

$$D = sc \cdot \sin\beta, \quad (3)$$

where β is the system tilt angle ($^\circ$) and sc is an empirically derived snow-

clearing coefficient. If no new snowfall occurs, the sliding conditions from the previous time step are retained. The sc value, originally defined empirically by Marion et al. as 0.197 for roof-mounted systems and 0.6 for ground-mounted systems [61], is handled differently in this work. Following the *Modified Marion* model [34], sc is determined by the snow depth on the PV system, defined as:

Table 3

Summary of different published original PV power simulation snow loss models.

Author(s)/year	Model Type	Model Methodology	Resolution
Powers et al. (2010) [32]	Power	Empirical	Monthly
Townsend and Powers (2011) [31]	Power	Empirical	Monthly
Andrews et al. (2013) [44]	Power	Empirical	Daily
Hong et al. (2013) [56]	Power	Empirical	Monthly
Zamo et al. (2014) [57]	Power	Empirical	Yearly
Shishavan et al. (2016) [58]	Power	Empirical	Monthly
Van Noord et al. (2017) [33]	Power	Empirical	Monthly
Ross (1995) [59]	Cover	First-principles	Intra-hour
Lorenz et al. (2011) [60]	Cover	Threshold	Intra-day
Marion et al. (2013) [61]	Cover	Threshold	Hourly
Bosman and Darling (2016) [62]	Cover	Threshold	Intra-day
Rahmatmand et al. (2016) [63]	Cover	First-principles	Intra-hour
Gun et al. (2018) [64]	Cover	Threshold	Hourly

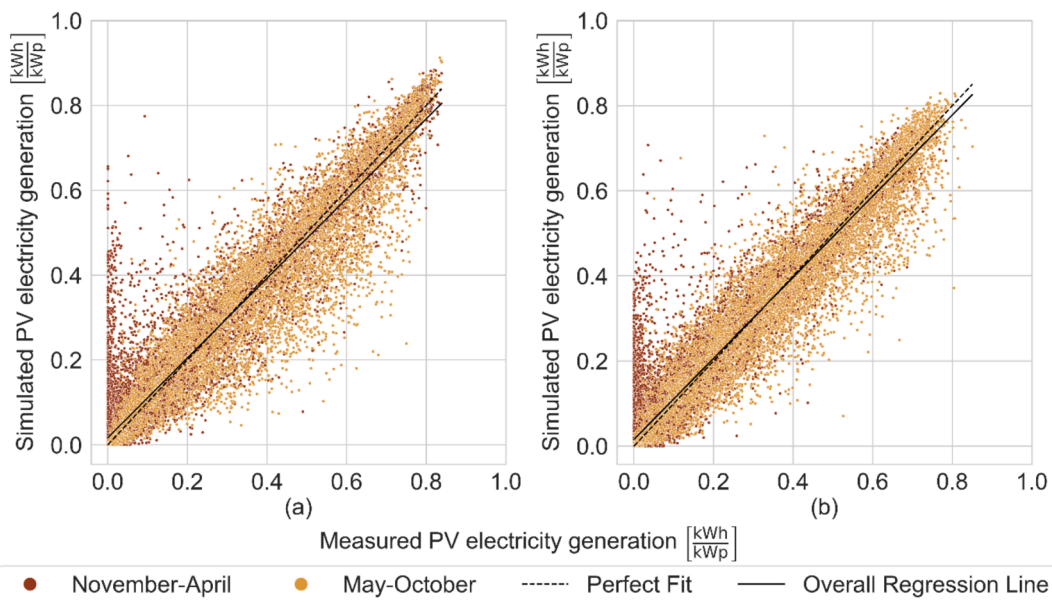


Fig. 4. Scatterplot of measured vs. simulated PV electricity generation for reference systems (a) R2 and (b) R18 for all available data for each respective system, using LiDAR-derived orientation, SARAH-3 irradiance data and Eq. (1) for power simulation. Each point represents one hour in the time series. The dotted line represents a perfect fit, the solid black line shows the fitted linear regression, and point colors indicate the time of year.

$$sc = \begin{cases} sc_1, SD_{PV,j} < 3cm \\ sc_2, SD_{PV,j} \geq 3cm \end{cases} \quad (4)$$

where the $SD_{PV,j}$ represents the cumulative snow depth on the modules at time step, j , and is calculated as:

$$SD_{PV,j} = \sum_{i=j_{PV,clear}}^j SF_i \cdot a_f \cdot \cos(\beta), \quad (5)$$

where $j_{PV,clear}$ is the last snow-free time step to the current time step, j . At each time step, i , snowfall, SF_i (cm), — hourly values in this study — is scaled by the accumulation factor, a_f , — set to 0.8 — to reflect slower snow buildup on roofs compared to the ground, due to wind redistribution and roof geometry. The cosine of the tilt angle further reduces accumulation by distributing the same amount of snowfall over a larger surface area compared to a horizontal surface.

2.5.2. Adjusting the Modified Marion model for remote sensing applications

The snow-depth dependent clearing coefficient captures increased melting and sliding for small snow depths and greater roof interference for larger depths. However, as noted by Øgaard et al. [34], snow clearing is system-dependent and influenced by technical factors such as submodule spacing [37], rear-side heating [59], and frameless module design [46], which may either enhance or hinder snow shedding. Consequently, additional parameter optimization was performed in this study, as detailed in Section 2.6.1.

Detailed data about system-specific characteristics are typically unavailable in remote sensing applications, limiting the feasibility of high-resolution snow loss modeling. Snow losses in this study are therefore estimated based on the fraction of the PV surface covered by snow, reducing absorbed irradiance. This departs from the original *Modified Marion* model, which calculates losses by identifying snow-covered parallel substrings [34], but improves applicability in data-limited contexts. Additionally, we use the previously described gridded baseline PV power simulation model rather than an IV-curve model (for reduced need of technical parameters and shorter computational time), meaning that only direct power output is simulated (not voltage and current), which further constrain the representation of detailed snow shading effects. This is because modeling only direct power output does not account for voltage and current behavior at the cell or string level, which determines how bypass diodes and inverters respond to partial snow coverage.

2.6. Snow loss model development and evaluation

Ideally, actual snow loss data would serve as the benchmark for evaluating model performance. However, estimating snow losses — especially at an hourly resolution — is challenging and introduces significant uncertainty. Therefore, a direct comparison of measured and simulated PV power generation, with and without the snow loss model, is used instead. This approach is valid given the quality-controlled reference data and the exclusion of heavily shaded systems (Section 2.2), which could otherwise confound snow loss estimates.

The evaluation metrics (Section 2.3) are calculated for filtered time series that only include:

1. Daytime hours (solar zenith angle $< 95^\circ$),¹ to exclude nighttime zero values which impact the metrics.
2. Periods when the snow loss model is active and/or snow depth is recorded in ERA5, accounting for regional snow variation.

¹ To capture low-angle irradiance during sunrise and sunset, a 5° buffer below the horizon was applied, corresponding to a zenith angle limit of $< 95^\circ$.

2.6.1. Model parameter optimization

To optimize the model, a maximizing function was implemented in Python, where the average of the MAE skill score, $\widehat{MAE}_{skill score}$, for the filtered time series with respect to all reference systems in the development set, was used as target parameter. The MAE skill score was optimized using a multi-parameter optimization, varying $TH_{snowfall}$, sc_1 and sc_2 at the same time. The maximization function is defined as:

$$\max \left(\widehat{MAE}_{skill score}(TH_{snowfall}, sc_1, sc_2) \begin{cases} 0 \leq sc_1 \leq 1.6 \\ 0 \leq sc_2 \leq 1.6 \\ 0 \leq TH_{snowfall} \leq 3.5cm \end{cases} \right). \quad (6)$$

2.6.2. Comparing the snow loss model with previous studies

As previously noted, the lack of reliable data on actual snow-induced losses presents a significant challenge when evaluating PV snow loss models. In this study we assess model performance by directly comparing hourly simulated PV output — including snow losses — with hourly measured generation data, which we consider the most viable evaluation method. To our knowledge, no prior snow loss studies have evaluated snow loss model performance using measured PV generation data at an hourly resolution. For example, Ryberg et al. [66] used monthly and annual resolution.

For consistency with existing literature, we also apply a complementary method based on Øgaard et al. [67] — hereafter referred to as the *Øgaard method* — which estimates snow losses from measured PV output and treats them as reference (“true”) values. Further methodological details are available in [34] and [68]. This approach enables consistent benchmarking of our snow loss model against a previous implementation of the *Modified Marion model*.

Our model is benchmarked against the data presented in Figure 5 of [65], which compares daily, monthly, and annual snow losses simulated with the *Modified Marion model* to losses estimated using the *Øgaard method*. Although our model operates at hourly resolution, this multi-temporal comparison provides a meaningful point of reference against recent work.

The main distinction in our adoption of the *Øgaard method* lies in the simulation setup: while the original approach uses pvlib [69] with the CEC IV-curve model to simulate snow-free PV power generation, and employs on-site irradiance and temperature measurements, we instead employ the simulation tool described in Section 2.4, relying on satellite- and reanalysis-based data from SARAH and MESAN.

Furthermore, a key difference in the snow loss estimation process in this study as compared to the *Øgaard method* [67], lies in the identification of snow loss days. Instead of assuming that all days with ground snow cover results in PV snow losses, we introduce an additional condition based on the Energy Performance Index (EPI) — defined as the ratio of measured to simulated PV power generation in [67]. A day is classified as a snow loss day only if ground snow cover is present and its EPI falls below the mean EPI of snow-free days minus 0.5.

This empirically chosen threshold aims to exclude days from the estimated snow loss data, which are unlikely to be affected by snow losses. The adjustment addresses the tendency to overestimate snow losses when relying solely on ground snow cover, as PV modules typically shed snow faster than the surrounding terrain due to their thermal and radiative properties.

2.6.3. Comparing Alfröduell with a previous suggested remote sensing pipeline

To provide context for the final simulation model results — including the enhancements described in the Appendix and the integration of the snow loss model, which together form the final simulation method in the *Alfröduell* remote sensing pipeline — we compare *Alfröduell* to a recently published state-of-the-art remote sensing and physical PV power simulation tool: *DeepPVMapper* [22]. Both frameworks employ comparable methods for detecting PV systems in aerial imagery, relying on image recognition and machine learning.

The *DeepPVMapper* method estimates PV system characteristics by segmenting aerial imagery and combining the results with national registry data for tilt, azimuth, and capacity. When orientation data are missing in the national registry, average regional tilt values are assumed, and DC capacity is inferred from polygon area and the assumed tilt. A standard module is applied in PVWatts, with fixed assumptions for AC/DC ratio (1.1) and inverter efficiency (96 %). In contrast, the *Alfröduell* pipeline derives module parameters based on installation year and average module parameters derived for that year from over 200 module datasheets to cope with historical module efficiency improvement trends. It uses a detailed inverter efficiency curve (Steca-grid 3000), an AC/DC ratio of 1.08 [70], and precise LiDAR-based estimates of tilt, azimuth, and effective area adjusted for by the derived module tilt [26].

In [22], the accuracy of simulated PV power generation from *DeepPVMapper* pipeline is evaluated using the percentage root mean square error (pRMSE), normalized by installed DC capacity. Results are presented for 255 systems in France under two scenarios: one using system characteristics estimated by the above-described methodology, and another using reference values from a system database. To enable a comparable assessment of both the quality of estimated system parameters and simulation accuracy of the two respective remote sensing and PV power simulations pipelines, we apply the same evaluation approach to our set of 40 reference systems.

3. Results and discussion

3.1. Snow loss model parameters optimization

The optimization results, based on data from the winters of 2021/2022, 2022/2023, and 2023/2024, revealed that the snowfall threshold ($TH_{snowfall}$) had the greatest impact on the snow loss model's MAE skill score. Fig. 5 illustrates the $\widehat{MAE}_{skill score}$ as a function of $TH_{snowfall}$. Each line represents a set of snow-clearing coefficient values. In this figure, only the case where $sc_1 = sc_2$ is displayed for demonstration purposes; however, the overall trend remains similar for other sc -value combinations.

Fig. 5 demonstrates that increasing the snowfall threshold $TH_{snowfall}$ from 0 cm leads to a substantial improvement across all configurations. However, beyond a threshold of 0.3 cm, the gains plateau, with most combinations of sc -values reaching their maximum improvement. Notably, very low threshold values cause the model to perform worse than the baseline for some sc -value combinations. These findings support the intended use of $TH_{snowfall}$, which primarily serves to filter out erroneous or negligible snowfall values from affecting the PV power output. In contrast to earlier implementations of the Marion model [53],

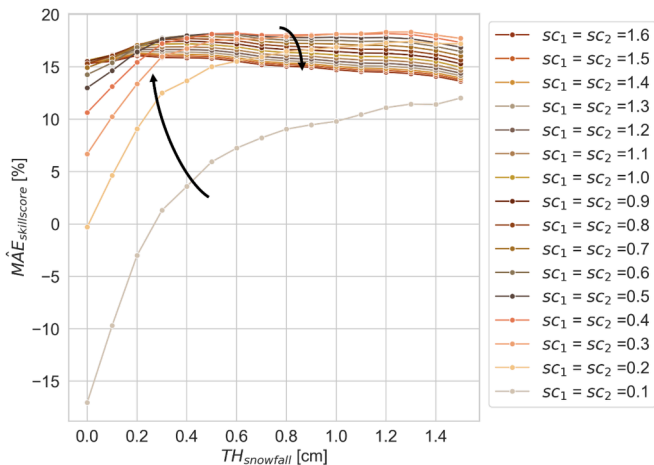


Fig. 5. $\widehat{MAE}_{skill score}$ improvement as a function of $TH_{snowfall}$. Each line represents a set of sc -values, and the arrows illustrate the trend with increasing sc -values.

which use a default $TH_{snowfall}$ of 1 cm, our findings suggest that lower $TH_{snowfall}$ values are better suited to our dataset.

Based on these findings, Fig. 6 visualizes the optimization results of the parameter space where $TH_{snowfall}$ is fixed at the value corresponding to the maximum observed improvement. The plot illustrates the average MAE skill score as a function of sc_1 and sc_2 , with the optimal parameter values for each system group summarized in Table 4.

Table 4 shows that the introduction of the snow loss model consistently results in approximately 2–4 % higher skill scores for tilted systems than for flat systems. The maximum $\widehat{MAE}_{skill score}$ are also accompanied by corresponding improvements in R^2 and RMSE, as expected due to their conceptual and mathematical relationship.

A consistent pattern across the two system groups is that optimal values occur at higher sc_1 than sc_2 , aligning with earlier findings in [34,67]. While these previous studies identified optimal values of $sc_1 = 0.3$, $sc_2 = 0.05$ for tilted residential systems and $sc_1 = 0.4$ and $sc_2 = 0.06$ for flat systems, our study yields generally higher values — though the same relative trend (lower values for tilted systems) is observed. This discrepancy between the studies may stem from differences in optimization strategies, as well as variations in system characteristics — such as heat leakage and roof geometry — that influence snow retention and removal.

Interestingly, contrary to the expectation that tilted systems would exhibit higher sc -values due to reduced roof interference, our results indicate the opposite. One possible explanation is that tilted systems experience more roof interference than expected — for instance, snow accumulating along module edges. Another contributing factor could be that the $\sin(\beta)$ -factor in Equation (3) has a greater impact at steeper angles.

Fig. 6 further supports these trends, showing similar patterns across both system types. Notably, a distinct ridge appears for the tilted systems between 0.2 and 0.4 for sc_2 — the sliding coefficient for thick snow layers (>3 cm) — which also contains the global maxima. This feature is less evident for flat systems, possibly because on tilted surfaces, thick snow layers tend to slide off abruptly once snow movement begins. Furthermore, the presence of the ridge indicates that when sc_2 is maintained between 0.2 and 0.4, sc_1 can vary from 0.5 to 1.5 with minimal impact on performance.

3.2. Snow loss model parameters validation

The resulting validation set skill scores from applying the optimal values of $TH_{snowfall}$, sc_1 and sc_2 — as determined in the model optimization and listed in Table 4 — are presented in Table 5 and visualized in Fig. 7. It should be noted that the validation set includes all available and quality-controlled power measurements recorded during the winter months (November–April) from 2017 to 2024 in various configurations.

Fig. 7 demonstrates that applying the optimized model parameters to the validation set of reference systems leads to improved performance across all evaluation metrics and system categories. This suggests that the derived parameters generalize well and support robust model performance on previously unseen systems, confirming that the snow loss model enhances the accuracy of PV power simulations in Sweden.

The lower RMSE skill score — shown in Table 5 and Fig. 7 — compared to the MAE skill score indicates that the model still struggles with outliers, particularly in cases where snow losses are overestimated (i.e., the model assumes snow cover when none is present). These instances produce large errors, dampening improvements in RMSE. While the MAE skill score improves consistently across all systems, both R^2 and RMSE skill scores show slight performance reductions for a few systems.

This variability is further illustrated in Fig. 8, which shows the weekly reduction in $\widehat{MAE}_{skill score}$ — where positive values indicate improved accuracy relative to the baseline model — achieved by the snow loss model over seven winter seasons using the validation dataset.

Fig. 8 shows that the weekly $\widehat{MAE}_{skill score}$ varies across systems, with

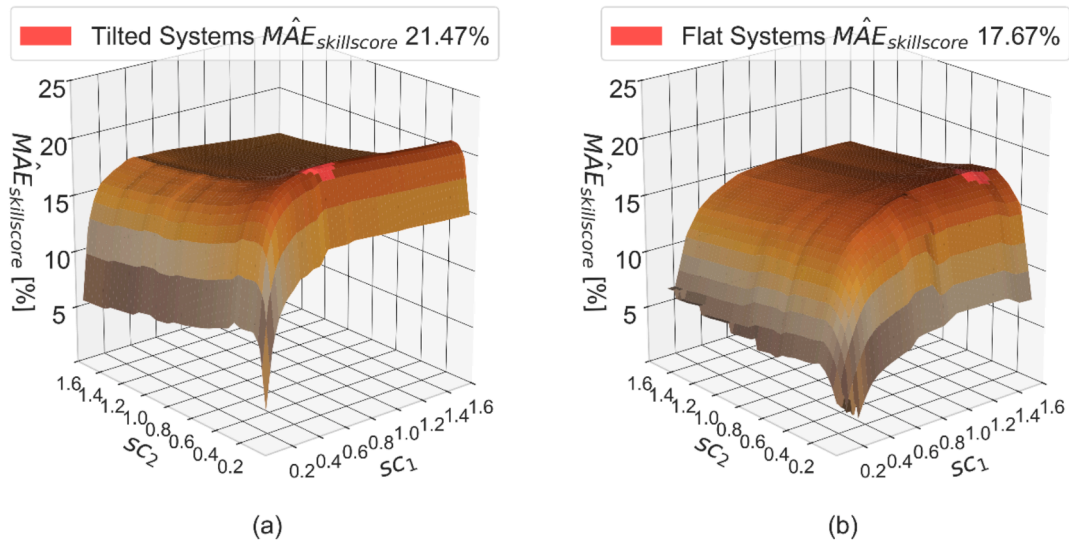


Fig. 6. 3D plot of the optimization using the development set of systems. (a) Shows the tilted systems, and (b) shows the flat systems in the development set. The red area indicates where the maximum is found. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Results of the snow loss mode parameter optimization, showing optimal values for each set of systems within the development set. The evaluation matrices are denoted with “hat”, indicating the average skill score increase for all development reference systems in each group as compared to the baseline simulations.

Category	$TH_{snowfall}$	sc_1	sc_2	$\hat{R}^2_{SkillScore}$	$\hat{RMSE}_{SkillScore}$	$\hat{MAE}_{SkillScore}$
All (16)	0.50	0.80	0.25	6.9 %	12.3 %	19.5 %
Tilted (11)	0.65	0.55	0.20	9.0 %	14.8 %	21.5 %
Flat (5)	0.50	1.40	0.40	5.1 %	13.0 %	17.7 %

systems R1 and R2 — located in Timrå and Gävle, respectively — exhibiting the most substantial improvements during winters with available data. These locations typically experience the highest snow-fall, which increases the likelihood for modeling errors due to snow accumulation. However, this was partly addressed during optimization by restricting skill score calculations to days with snow cover or simulated snow loss. It can be noted that the $\hat{MAE}_{SkillScore}$ reach 100 % for several systems over multiple weeks. This indicates that these systems are fully covered by snow, and that the snow loss model accurately simulates the snow cover during these periods.

Overall, the average weekly $\hat{MAE}_{SkillScore}$ remains positive, supporting the conclusion that the snow loss model improves simulation accuracy on average. Nonetheless, some systems display periods of increased errors — most notably systems R11 in Nynäshamn, which show increased errors across multiple weeks. In these cases, the model struggles to accurately capture snow loss dynamics, highlighting limitations in specific regional or site conditions.

Table 5

Validation set results, using the overall (all system) optimal model parameters from Table 4. The evaluation matrices are denoted with “hat”, indicating the average skill score increase for all validation reference compared to the baseline simulations.

Category	$TH_{snowfall}$	sc_1	sc_2	$\hat{R}^2_{SkillScore}$	$\hat{RMSE}_{SkillScore}$	$\hat{MAE}_{SkillScore}$
All (9)	0.50	0.80	0.25	6.3 %	15.7 %	21.7 %

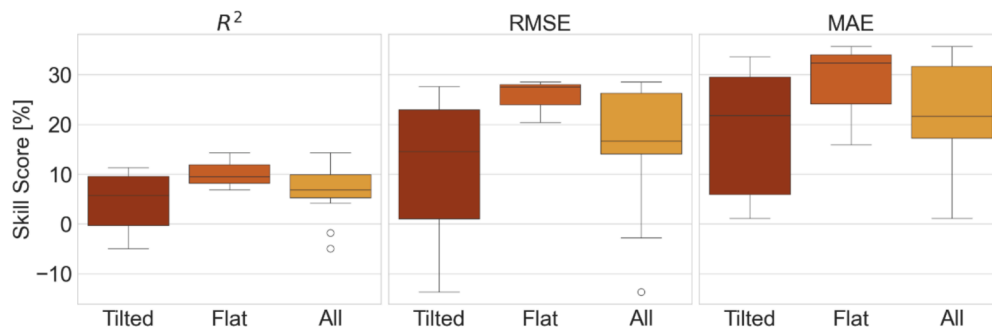


Fig. 7. Boxplots of skill scores for the validation set systems, using the optimal model parameters from the optimization procedure, are shown separately for tilted and flat PV systems, as well as for all validation systems combined.

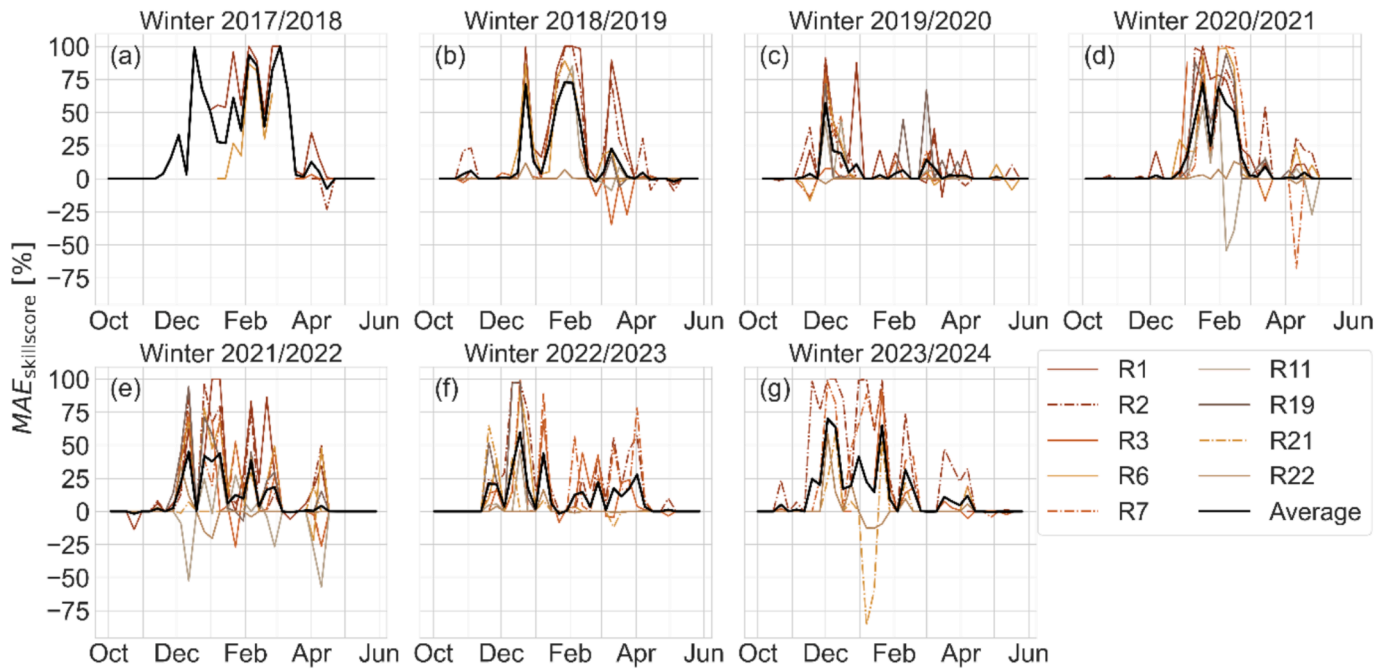


Fig. 8. Weekly $MAESkillscore$ compared to the baseline, using the snow loss model with optimized parameters applied to the validation set of reference systems. Positive values indicate improved performance. The black line represents the average weekly $MAESkillscore$ across all systems with available data for each winter. Flat residential systems are shown with dashed lines, while tilted systems are depicted with solid lines. Subplots (a)–(g) correspond to winter seasons from 2017/2018 to 2023/2024, respectively.

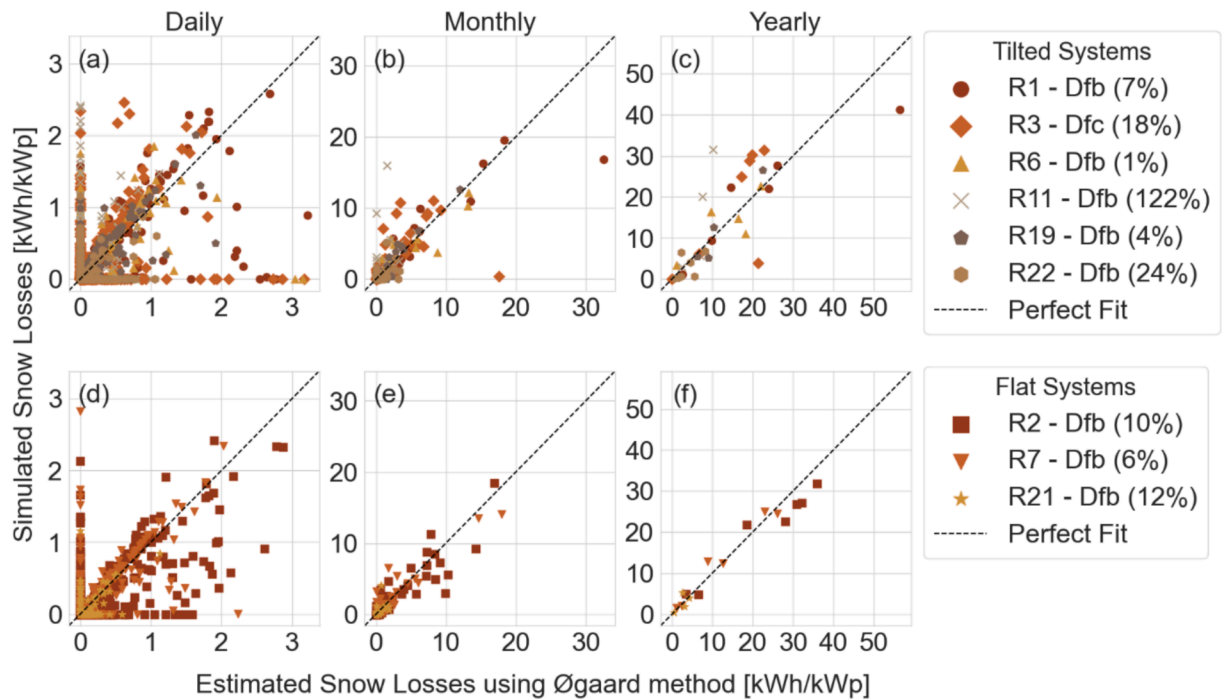


Fig. 9. The daily, monthly and yearly simulated snow losses compared to estimated snow losses derived from measured PV power generation data, using the Øgaard method of [67]. The top row shows the tilted systems in the validation set, with subfigures (a)–(c) corresponding to daily, monthly, and yearly resolutions, respectively. The bottom row displays the flat systems in the validation set, with subfigures (d)–(f) representing daily, monthly, and yearly resolutions, respectively. The legend shows the total absolute error between simulated snow loss and estimates derived using the Øgaard method, indicated in parentheses.

3.3. Comparison with a previous application of the Modified Marion model

In this section, the implemented snow loss model is benchmarked against previous applications of the *Modified Marion* model —

specifically [67]. Simulated snow losses for the reference systems in the validation set — generated using the optimized model parameters from Section 3.1 — are compared with estimated snow losses derived using the *Øgaard method* [67] for the same systems (see Section 2.6.2).

Fig. 9 presents a comparison between simulated snow losses and

those estimated using the *Øgaard method*, inspired by the layout of Figure 5 in [67]. It also shows the total absolute snow loss error, relative to the *Øgaard method* estimates based on measured generation data, to serve as a benchmark for evaluating the snow loss model.

Fig. 9 shows high agreement between estimated snow losses by the *Øgaard method* and simulated snow losses at both yearly and monthly resolutions. At the daily level, however, the correlation weakens: while many points align with the 1:1 line, others cluster along the x- and y-axes. It is important to note that the estimated snow losses using the *Øgaard method* [67] are approximations, subject to uncertainty, and do not exactly represent actual snow losses. Nonetheless, they are expected to capture most snow loss events. Therefore, not all points along the y-axis necessarily reflect snow loss model errors, nor do all points along the x-axis indicate missed snow loss events by the *Øgaard method*. Overall, Fig. 9 suggests that the model aligns well with estimated losses by the *Øgaard method* on many days but does not capture all snow losses. Furthermore, comparing Fig. 9 to Figure 5 of [67] reveals a strong agreement across daily, monthly and yearly timescales, suggesting that our simplified snow loss model performs similarly to the original *Modified Marion model*.

Comparing the total absolute snow loss error with Figure 5 in the study of [67], presented values in the range of 3 % to 46 %, and an average error of 18 % over 10 systems. Our results show errors in the range of 1–112 %, with an average of 21 % over 9 systems, with all except one system with modelled error below 25 %. The outlier, system R11, is exhibiting a significantly larger absolute total snow loss error compared to the other systems, further reinforcing the observed decrease in accuracy relative to the baseline simulation, as illustrated in Fig. 8. One possible explanation for this discrepancy is increased roof interference or unexpected heat leakage from the building — factors that the model does not explicitly account for. Additionally, inaccuracies in the weather and snowfall data for this system, or potential inconsistencies in the measured PV power generation data — despite quality control measures — could also contribute to the observed modeling errors.

3.4. Final PV power simulation results

After implementing the developed snow loss model to the PV power simulation tool, simulations were conducted for all 40 reference systems using available data for each system. This set included the 15 systems previously excluded from snow loss model development and validation due to suspected shading from nearby obstacles. The results are presented in Fig. 11, with complementary data provided in Table 6.

The final simulation results indicate that the snow loss model has a relatively small effect on the overall performance of the simulation model. However, as illustrated in Fig. 10, the model improves simulation accuracy across all evaluation metrics. Specifically, the mean R^2 value increases from 0.888 to 0.894, the MAE decreases from 0.037 to 0.036 kWh/kW_p, and the RMSE decreases from 0.066 to 0.064 kWh/kW_p, as shown in Table 6.

The limited impact on overall model performance is expected, as snow-related losses typically account for only 2–10 % of annual power losses in northern and central Sweden (Dfc) [33], while most reference systems are located in central and southern Sweden (Dfb). Nonetheless,

the model captures key snow loss dynamics and, in some cases, fully explains observed losses. This is illustrated in Fig. 8, where the MAE skill score reaches 100 % for certain systems during specific periods.

Fig. 11 shows that the unshaded reference systems that were used for development and validation exhibit high accuracy, with R^2 values between 0.88–0.96, and low MAE and RMSE values ranging from 0.02 to 0.04 and 0.04–0.08, respectively. For the shaded systems, which were excluded from the snow loss model development, R^2 , MAE, and RMSE values vary between 0.67 and 0.93, 0.03–0.06, and 0.05–0.11, respectively. This indicates that the Alfröduell remote sensing and PV power simulation pipeline demonstrates high performance for most distributed systems, while performance is reduced for shaded systems.

While the Alfröduell remote sensing and PV power simulation pipeline demonstrates good overall accuracy — particularly given that all technical input data are derived from remote sensing sources only — Fig. 11 also demonstrate that its performance decreases for systems affected by shading. This suggests that the pipeline would benefit from the integration of a shadow loss model to better account for the effects of nearby obstacles. Incorporating such a model could improve accuracy for shaded systems and further enhance the overall performance of the Alfröduell pipeline across spatially distributed PV systems. This enhancement is also feasible as LiDAR data, which is already used in the Alfröduell pipeline, can be used to classify, measure and localize shadowing objects [29,71,72], a prerequisite for effective and accurate shadowing modelling.

In our model evaluation framework, simulated PV power generation is compared to measured data to assess the snow loss model's effectiveness in reducing errors relative to a baseline without snow loss modeling, using MAE as the metric. However, this approach inherently attributes all error reductions to the snow loss model. In practice, other factors — such as uncertainties in irradiance data or unmodeled shading — can produce deviations between simulated and measured PV output that are indistinguishable from snow-related losses in the MAE metric. Consequently, some of the observed error reduction may be due to these unrelated factors rather than snow loss alone. Considering that systems with significant shading were excluded, and the simulation model's MAE remains within 0.02–0.04 kWh/kW_p at an hourly resolution, we judge the overestimation to be small.

An alternative would be to optimize the model by directly comparing simulated and estimated snow losses from the *Øgaard method*, i.e., tuning model parameters by comparing simulated and estimated losses rather than simulated and measured power. This could offer insights into the proportion of correctly identified snow-loss days and the magnitude of modeling errors. However, snow losses are difficult to detect in PV power data, as other winter losses — such as low irradiance and large angles of incidence — may be misclassified. Inaccurate snow-loss estimates could therefore introduce further uncertainty, leading to optimization based on non-snow-related events and reducing model accuracy. Given these considerations, we believe our approach remains preferable to direct optimization against estimated snow losses derived from PV data.

Furthermore, the model does not account for time-delayed effects — such as a cold night freezing the bottom snow layer to the PV array [67] — potentially leading to an underestimation of snow retention duration.

Compared to previous implementations of the *Modified Marion model* [67], our results show similar trends in the relationship between simulated and estimated losses across all time resolutions and system types. These results indicate that our snow loss model performs on par with previous studies, despite necessary simplifications in PV power simulation and snow loss modeling to accommodate the limited data available through the remote sensing approach.

The limitations of the developed snow loss model include sensitivity to rare snow events and exclusion of wind-driven clearing effects. Although wind speed data are available, it is not incorporated into the *Modified Marion model*. Incorporating wind effects into the modeling of snow accumulation and clearing processes may better capture the

Table 6

Performance metrics of the Alfröduell PV power simulation pipeline are presented for all 40 reference systems, using database derived system parameters.

Model & Metric	Min	Max	Mean	Median
Baseline – R^2	0.643	0.956	0.888	0.913
Snow – R^2	0.667	0.956	0.894	0.918
Baseline – MAE	0.025	0.062	0.037	0.036
Snow – MAE	0.025	0.060	0.036	0.034
Baseline – RMSE	0.041	0.109	0.066	0.066
Snow – RMSE	0.041	0.105	0.064	0.062

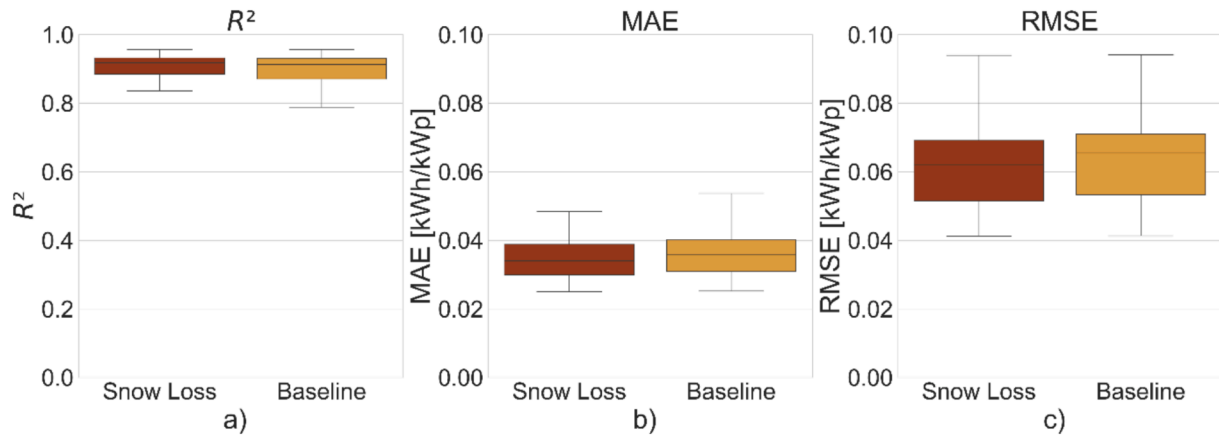


Fig. 10. Boxplot of performance metrics for all 40 reference systems, including the shaded systems that were excluded from the main analysis. Subfigures (a), (b), and (c) show the matrices for R^2 , MAE, and RMSE, respectively. Results from simulations incorporating the snow loss model are shown in red, while baseline simulation results are shown in yellow. Simulations were performed using all available data for each system, excluding nighttime hours. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

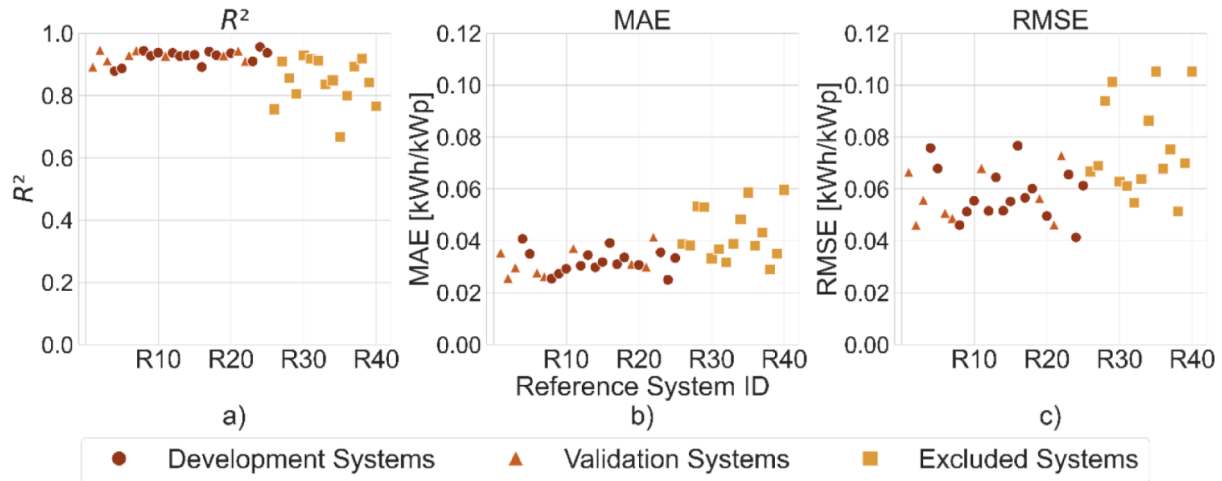


Fig. 11. Performance metrics of the Alfrödull PV power simulation pipeline are presented for all 40 reference systems, using database system parameters and incorporating the snow loss model. Simulations were run using only daytime hourly data for each system. Circles indicate the 16 reference systems used for developing the snow loss model, triangles represent the nine systems used for validation, and squares denote the 15 systems excluded from model development due to suspected shading from nearby obstacles. Subfigures (a), (b), and (c) show the results for R^2 , MAE, and RMSE, respectively.

dynamics of snow accumulation on PV panels [31,35,67], particularly during high wind events [45,59], and could potentially enhance the accuracy of the snow loss model [66]. However, the implementation of wind effects into a snow loss model would require extensive analysis of historical data, since the properties of a snowpack are an effect of the historical weather events and conditions. For example, a snowpack that has consolidated during a warm spell would likely not be affected by strong winds, whereas cold low-density snow would. Furthermore,

Table 7

Comparison of simulation accuracy (pRMSE [%]) between the DeepPVMapper [22] and Alfrödull remote sensing pipelines using both database and remotely sensed derived PV system characteristics.

	Min	Max	Mean	Median	N
DeepPVMapper – Database	3.90 %	26.49 %	8.36 %	7.66 %	255
DeepPVMapper – Remote sensing pipeline	4.15 %	43.39 %	10.10 %	8.18 %	255
Alfrödull – Database	3.29 %	8.51 %	5.23 %	4.95 %	40
Alfrödull – Remote sensing pipeline	3.39 %	9.43 %	5.71 %	5.46 %	40

variations in the relationship between absorbed and transmitted irradiance through a snowpack and ambient temperature, as presented in [43], can be paired with *Modified Marion* model. These two options — incorporating wind effects and the inclusion of absorbed irradiance and thermal capacitance — present opportunities for future model refinements that could improve performance without compromising the remote sensing-based approach.

3.5. Comparison of the Alfrödull and DeepPVMapper remote sensing pipelines

The comparative results between the Alfrödull and DeepPVMapper remote sensing and PV power simulations pipelines are summarized in Table 7, where the ‘Alfrödull – Database’ configuration corresponds to the setup implemented to generate the results shown in Fig. 11, but with the addition of nighttime hours, which are included in the results reported in [22], to enable a fair comparison.

As shown in Table 7, the Alfrödull method consistently yields lower pRMSE values across all metrics compared to DeepPVMapper. Notably, Alfrödull’s performance using remotely sensed system parameters

closely matches the database values. This highlights the high accuracy of the LiDAR-based system characterization, particularly in estimating tilt and azimuth. The two remote sensing pipelines were evaluated on two distinct reference sets in different regions. While using a single reference set would have been more informative, this was not feasible; hence, the results should be considered an initial benchmark rather than a fully comparable evaluation.

Our results showcase *Alfrödull*'s ability to accurately estimate PV power output, even under data-limited conditions. Importantly, the integration of the snow loss model developed in this study makes *Alfrödull* particularly well-suited for regions like Sweden and other high latitude and altitude regions where snow accumulation can substantially impact PV performance.

4. Conclusions

In this study we develop, optimize, and validate a modified version of the Marion snow loss model for PV systems, adapted for remote sensing applications. The model was optimized using data from 16 PV reference systems (both flat and tilted), and validated on an additional nine, all located in Sweden. It demonstrates improved skill scores across all evaluation metrics and shows good generalizability. The snow loss model accuracy matches that of previous modified Marion model adaptations, while relying solely on remotely sensed inputs such as satellite-derived irradiance and weather data, along with LiDAR-based azimuth and tilt of the PV systems.

The snow loss model is integrated into the *Alfrödull* remote sensing and PV power simulation pipeline, where the PV power simulation component is based on an improved version of a previously validated gridded PV power simulation model. Additionally, enhancements to the temperature estimation and wind correction models, combined with the use of SARAH-3 irradiance data, led to substantial improvements in accuracy. With these improvements, and the integration of the snow loss model into the *Alfrödull* remote sensing and PV power simulation pipeline, R^2 values of 0.67–0.96 and MAE scores of 0.02–0.06 across 40 reference PV systems were achieved.

This implies that snow losses can be accurately modeled using only generic PV module parameters and remote sensing data as input, enabling site-specific PV system generation assessments at scale in cold climate zones.

Appendix

PV power model comparison

Several physical models exist for simulating photovoltaic (PV) power output, including the widely used approach of modeling the current–voltage (IV) curve with the one-diode equation. As a simplified alternative, a gridded simulation model — originally based on the linear physical PV power generation model introduced in [54] — was developed by Campana et al. [55] for applications in Nordic climates. The two PV power models are here compared across simulations from eight Swedish reference systems, including data from the seven CheckWatt reference systems [21] and one Swedish PV park [73] in different parts of the Sweden. The IV curve was modelled by solving the one-diode equation using the De Soto et al. model [74], requiring computation of module cell temperature, which was done with the Sandia Array Performance Model [75]. The Sandia Grid-Connected PV Inverter model [75] was used for the IV model, while the gridded model employed the Steca Grid 3000 efficiency curve.

Comparison of the 11 simulations (7 for the distributed systems over several years and 4 monthly simulations for the PV park) with the measured PV output showed that both models had high R^2 -values ranging between 0.68 and 0.84 for the rooftop systems using STRÅNG irradiance data, and 0.92–0.99 for the PV park with on site-measurements irradiance measurements, indicating good overall performance. Comparing the two models revealed a minimal difference in R^2 -values, with the gridded model exhibiting an on average slightly higher R^2 -value of 0.849 as compared to 0.848 for the one-diode equation. Although the gridded model showed slightly better accuracy in some cases, the IV curve model generally demonstrated slightly better accuracy and precision based on lower values of both MAE and RMSE values closer to zero.

Overall, the differences in performance between the two models are relatively small and the choice depends on specific priorities and model application. It should be noted that the Sandia inverter model used in the IV model relies on detailed parameters generally not provided by manufacturers in the standard data sheets. Its applicability hinges on access to inverter model parameters within the System Advisor Model database [76].

In addition to the accessibility of product specific data, a computational time analysis showed a significant difference between the runtimes of the two models (p-value of 0.0), simulating the PV power output ($N = 214742$) for a single system orientation. From 1000 simulations, the Gridded optimization model exhibited a mean runtime of 0.0038 s with a standard deviation of 0.0007, while the PV IV-curve model's mean runtime was considerably higher at 1.8985 s and with a standard deviation of 0.0476. The considerably longer computational time for the IV-curve model is a

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the article. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRediT authorship contribution statement

Gustav Öhgren: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lisa Molin:** Writing – review & editing, Software, Formal analysis, Data curation. **Mattias Lindh:** Writing – review & editing, Validation, Supervision. **David Lingfors:** Writing – review & editing, Supervision. **Johan Lindahl:** Writing – review & editing, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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ERA5-Land data from Muñoz-Sabater et al. [49] were obtained via the Copernicus Climate Change Service (2024). The results contain modified Copernicus Climate Change Service information (2021). Neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains.

major drawback for the intended application, which is to simulate power generation from a large number of individual PV systems within a geographical area for several years. Consequently, in accordance with the findings of [55], the performance between the two models are relatively small and since the simplified gridded model offers greater flexibility, due to its simplicity — in terms of needed technical parameters — and shorter computational time, the gridded optimization model is the more favorable option for the intended application and is therefore used as the baseline PV power simulation method.

Improvements of the gridded PV power simulation model

Earlier versions of the gridded PV power model [21,55] estimated cell temperature by using equation 23.3.4 from [54], which assumes nominal operating cell temperature (NOCT) conditions. However, NOCT is based on several assumptions, including free-standing modules oriented towards solar noon with equal exposure on both sides [77] — conditions that are rarely met by rooftop PV systems, where modules often experience asymmetric temperature environments. Given that rooftop installations typically deviate from NOCT conditions, the previous temperature model used in the gridded optimization framework is in this study replaced by a modified version of Equation 10.37 from [78]:

$$T_c = T_a + \frac{G_{g,t}}{800 \text{ W/m}^2} (T_{NOCTa} - 20^\circ\text{C}) \left(1 - \frac{\eta}{\tau\alpha}\right) \frac{9.5 \text{ W/m}^2\text{C}}{5.7 \text{ W/m}^2\text{C} + 3.8 \text{ W/m}^3\text{C} \cdot \text{WS}_{PV}}, \quad (\text{A.1})$$

where T_c is the cell temperature ($^\circ\text{C}$), T_a is the ambient temperature ($^\circ\text{C}$), $G_{g,t}$ is the incoming global tilted irradiance (W/m^2), T_{NOCTa} is the NOCT temperature ($^\circ\text{C}$) adjusted based on the system mounting according to equation 10.36 in [78], η is the efficiency of the PV module at STC, $\tau\alpha$ is the transmittance absorptance product, and WS_{PV} is the wind speed (m/s) at the height of the PV system.

In [21], the simulation model was validated using the original MESAN wind speed data [47] calculated at 10 m above ground level (WS_{10m}), and applied directly in PV system simulations. To improve the accuracy of the wind speed input at the PV system level, building-specific installation heights derived from LiDAR data (via the third step of the *Alfröduell* remote sensing pipeline) are used to correct these values using the logarithmic wind profile law [79]:

$$\text{WS}_{PV} = \text{WS}_{10m} \left(\frac{\ln(z_{PV}/z_0)}{\ln(z_{10m}/z_0)} \right), \quad (\text{A.2})$$

where WS_{PV} is the corrected wind speed (m/s) at the PV system height z_{PV} (m), z_{10m} is the reference height of 10 m and z_0 is the terrain-dependent roughness length (m). The value of z_0 is estimated using standard values based on the geographical location of the system. Compared to the previous temperature model and wind speed assumption, Eq. (A.1) typically yields higher temperature estimates and by implementing this model the RMSE skill score was increased by 1.3 % and the MAE skill score by 1.8 % for our reference PV systems simulations. The updated temperature model was further validated at two test sites using on-site weather and irradiance data — one ground-mounted and one roof-mounted — where temperature calculations were based on NOCT values adjusted to each system's mounting configuration, which provided the best model performance [80]. In this evaluation it achieved MAE skill score improvements of 6.8 % (tilted roof) and 1.3 % (ground).

The updated equation for simulating the PV power generation per installed module capacity ($P_{DC/kW}$) in this study is hence as follows:

$$P_{DC/kW} = \frac{S \cdot \eta \cdot \left(1 + \frac{\mu_{Voc}}{V_{mp}} \left(T_a - T_{STC} + \frac{G_{g,t}}{800} (T_{NOCTa} - 20) \left(1 - \frac{\eta}{\tau\alpha}\right) \frac{9.5}{5.7 + 3.8 \text{WS}_{PV}}\right)\right)}{G_{STC} \cdot \eta}, \quad (\text{A.3})$$

with same units for the constants as in Eq. (A.1) and where S is the PV system effective absorbed global tilted irradiance (W/m^2), μ_{Voc} is the open circuit voltage temperature coefficient ($\text{V}/^\circ\text{C}$), V_{mp} is the voltage at maximum power point (V), T_{STC} is the temperature during STC (25°C) and G_{STC} is the global tilted irradiance at standard test conditions (STC), which is equal to 1000 W/m^2 . The ratio between S and $G_{g,t}$ is used to estimate $\tau\alpha$.

While a previous validation study [21] relied on irradiance data from STRÅNG [81], this study employs irradiance data from SARA-3 [48] due to its enhanced accuracy. Simulating the power generation for all 40 reference systems (including shaded system) with the two irradiance data sources, respectively, showed that the model improved with the SARA-3 irradiance data with an average R^2 skill score increase of 10.6 % and a MAE skill score increase of 28.9 %. These results are consistent with Campana et al. [55], who found SARA-2 to outperform other irradiance models when validated against irradiance measured at Swedish weather stations.

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